

# tf.compat.v1.sparse\_add Stay organized with collections Save and categorize content based on your preferences.

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*Adds two tensors, at least one of each is a SparseTensor. (deprecated arguments)*

## Function Signatures

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```
tf.compat.v1.sparse_add( a, b, threshold=None, thresh=None )  
(thresh)
```

## Code Examples

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```
tf.compat.v1.sparse_add(  
a, b, threshold=None, thresh=None  
)
```

```
tf.compat.v1.sparse_add(  
a, b, threshold=None, thresh=None  
)
```

```
[      2]  
[.1     0]  
[ 6    -.2]
```

```
[      2]  
[.1     0]  
[ 6    -.2]
```

## Documentation

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Adds two tensors, at least one of each is a SparseTensor. (deprecated arguments)

### View aliases

Compat aliases for migration

See

Migration guide for  
more details.

`tf.compat.v1.sparse.add`

If one `SparseTensor` and one `Tensor` are passed in, returns a `Tensor`. If both arguments are `SparseTensors`, this returns a `SparseTensor`. The order of arguments does not matter. Use vanilla `tf.add()` for adding two dense `Tensors`.

The shapes of the two operands must match: broadcasting is not supported.

The indices of any input `SparseTensor` are assumed ordered in standard lexicographic order. If this is not the case, before this step run `SparseReorder` to restore index ordering.

If both arguments are sparse, we perform "clipping" as follows. By default, if two values sum to zero at some index, the output `SparseTensor` would still include that particular location in its index, storing a zero in the corresponding value slot. To override this, callers can specify `thresh`, indicating that if the sum has a magnitude strictly smaller than `thresh`, its corresponding value and index would then not be included. In particular, `thresh == 0.0` (default) means everything is kept and actual thresholding happens only for a positive value.

For example, suppose the logical sum of two sparse operands is (densified):

- `thresh == 0` (the default): all 5 index/value pairs will be returned.
- `thresh == 0.11`: only .1 and 0 will vanish, and the remaining three

index/value pairs will be returned.

- `thresh == 0.21`: .1, 0, and -.2 will vanish.

## Returns

`## Raises`

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